Available from October 2026
sacha.durain@gmail.com
+33 6 35 02 05 94
linkedin.com/in/sacha-durain
sdurain.github.io

Sacha Durain

Future PhD Graduate – Computational and Structural Dynamics

Hybrid profile in computational mechanics and artificial intelligence, specialized in the development of digital twins and surrogate models from experimental data, finite element simulations, and dynamic signals.

Education and Academic Experience

Future PhD Graduate – Computational and Structural Dynamics 2023 – present

LaMcube & LAMIH – University of Lille | Lille & Valenciennes, France

- *Dissertation: "Friction-induced vibrations: experiments and modeling of triggering conditions through enriched interface parameters." Supervised by Prof. P. Dufrénoy and Prof. F. Massa.*
- Acquired, structured, and processed instrumented test data (multimodal sensors) and numerical simulations coupled with reduced-order models.
- Developed surrogate models under uncertainty for a braking-system digital twin: CNNs, MLPs, Gaussian processes, recurrent models, deep ensembles, Monte Carlo methods, and SHAP-based interpretability.
- Semantic segmentation of brake-pad material images (SEM and optical) with unsupervised pseudo-labeling and transfer learning (Transformers).
- Disseminated the work through 3 scientific presentations and 5 published or submitted journal articles.

Teaching Assistant 2023 – present

Polytech Lille & INSA Hauts-de-France | Lille & Valenciennes, France

- *Dynamics & signal processing:* taught system dynamics and fundamentals of signal processing.
- *MATLAB programming:* supervised student projects in finite element simulation, optimization, and data analysis.

Engineering Degree – Mechanical Simulation 2018 – 2023

INSA Hauts-de-France | Valenciennes, France

- Core coursework: optimization, finite element methods, programming, and structural dynamics.

Industrial Experience

Simulation & Design Engineer – Apprentice 2020 – 2023

Alstom | Petite-Forêt, France

- Designed and validated railway products through mechanical simulation and data analysis.
- Collaborated with multidisciplinary and international teams across several sites.

Mechanical Engineering Intern May 2021 – August 2021

Canray | Bursa, Turkey

- Conducted test-simulation correlations on automotive components.
- Supported quality monitoring and technical exchanges with the client.

Core skills

Deep learning & machine learning: PyTorch, scikit-learn **Numerical simulation**: Ansys, Abaqus, HyperMesh
Data analysis & visualization: NumPy, pandas, Matplotlib, Plotly
Programming: Python, MATLAB, Bash, HPC, Slurm, Git **Languages**: French (native), English (C1)

Academic Referees

Prof. Franck Massa (INSA Hauts-de-France & UPHF): franck.massa@uphf.fr
Prof. Philippe Dufrénoy (University of Lille): philippe.dufrenoy@univ-lille.fr

Research output

Publications

1. **S. Durain**, Q. Caradec, M. Briatte, J.-F. Brunel, C. Hubert, F. Massa, P. Dufrénoy.

   Enhancing the prediction of squeal through multiphysic and multiscale analysis of experimental data from a pin-on-disk system.

   Mechanical Systems and Signal Processing, vol. 247, p. 113968, 2026.

   doi:10.1016/j.ymssp.2026.113968

2. Q. Caradec, **S. Durain**, M. Thévenot, M. Briatte, M. Stender, J.-F. Brunel, P. Dufrénoy.

   Pin-on-Disc Experimental Study of Thermomechanical Processes Related to Squeal Occurrence.

   Lubricants, vol. 13, no. 4, art. 186, 2025.

   doi:10.3390/lubricants13040186

3. **S. Durain**, Q. Caradec, M. Briatte, J.-F. Brunel, C. Hubert, F. Massa, P. Dufrénoy.

   Experimental–numerical validation of a contact-interface-induced squeal model under uncertainty.

   Preprint (*Mechanical Systems and Signal Processing*).

4. **S. Durain**, Q. Caradec, M. Briatte, J.-F. Brunel, C. Hubert, F. Massa, P. Dufrénoy.

   A numerical framework for long-time friction-induced vibration simulations based on an enhanced contact description.

   Preprint (*Journal of Sound and Vibration*).

5. **S. Durain**, J.-F. Brunel, C. Hubert, F. Massa, P. Dufrénoy.

   Evolution and impact of waviness during friction test.

   Preprint (*Tribology International*).

Conference presentations

1. **S. Durain**, Q. Caradec, M. Briatte, J.-F. Brunel, C. Hubert, F. Massa, P. Dufrénoy.

   Enhanced experimental–numerical correlation for the investigation of dynamic instabilities under uncertain frictional interface modeling.

   17th National Colloquium on Structural Computation (CSMA 2026), Presqu'île de Giens, Var, France, 18–22 May 2026.

2. **S. Durain**, Q. Caradec, M. Briatte, J.-F. Brunel, C. Hubert, F. Massa, P. Dufrénoy.

   Enrichment of the contact-interface description for the identification of friction-induced dynamic instabilities.

   26th French Congress of Mechanics (CFM 2025), Metz, France, 25–29 August 2025.

3. Q. Caradec, **S. Durain**, M. Thévenot, M. Briatte, M. Stender, J.-F. Brunel, P. Dufrénoy.

   Experimental and numerical study of the thermomechanical and tribological mechanisms involved in dry friction brake emissions.

   European Solid Mechanics Conference (ESMC 2025), Lyon, France, 7–11 July 2025.